Algebra in Action:
Adventures with Anna

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1. The A4 lectures and notes

On 3rd September 2026, I will give a couple of lectures on the topic of 'Algebra in Action: Adventures with Anna' (A4). The abstract is:

Abstract. Across scientific domains, researchers seek to understand complex systems by decomposing them into simpler parts. These lectures will discuss how to build a theoretical and practical framework for such problems, using the tools and insights of metric algebraic geometry. I will also report on enjoyable conversations and collaborations to study real-world systems. The goal is to initiate a conversation on open questions and future directions.

These (evolving) notes complement the lectures. Their goal, for now, is to help prepare the audience. The adventure will be in three stages:

(1) the paved trail: classical data analysis methods and the algebra behind them

(2) the hiking trail: non-standard and present-day methods

(3) going off-trail: ideas for the future; into the unknown

For even more background, the lecture notes for (the first half of) my graduate course 'Algebraic fundamental of representing data' are available at https://seigal.github.io/module1.pdf and https://seigal.github.io/module2.pdf. Another short related reference is [10].

Comments on all are welcome. See you at A4.

Anna

2. Introduction

Algebra in action means algebra for mathematical data science: algebraic ideas designed or applied with the goal of answering a data analysis question. Examples of such questions may include those in biology, literary studies, social sciences, etc. The questions have both trans-disciplinary and domain-specific aspects. We will reflect on what makes a data analysis question amenable, or not, to an algebraic point of view.

A central idea in the study of complex systems, such as those of the above applications, is the hypothesis of compositionality: observed variables (typically high-dimensional) can be described as combinations of (usually unobserved) simpler parts.

Formalized via random vectors, this hypothesis writes an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as $\mathbf{x} = f(\mathbf{z})$ for a latent random vector $\mathbf{z} \in \mathbb{R}^q$, and a fixed unknown function $f : \mathbb{R}^q \to \mathbb{R}^p$. The special case where f is linear replaces it by a fixed, unknown matrix $A \in \mathbb{R}^{p \times q}$. We seek to recover f or A and properties of $\mathbf{z}$ from samples of $\mathbf{x}$. This is a factorization problem. Many ideas in data analysis can be thought of as factorization problems. We begin with the earliest, most classical examples.

3. The paved trail

3.1. What is a factor? Multivariate data is a collection of vectors $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)} \in \mathbb{R}^p$. Each data vector records measurements of p variables. Such data can be organized into a matrix $X \in \mathbb{R}^{n \times p}$: the n rows are the samples, the p columns index the variables, and entry X_{ij} is the value of variable j in sample i. For example:

Sample i	Variable j	x_{ij}
cell	gene	expression level of gene j in cell i
timepoint	electrode	voltage of electrode j at time i
image	pixel	intensity of pixel j in image i
document	word	count of word j in document i

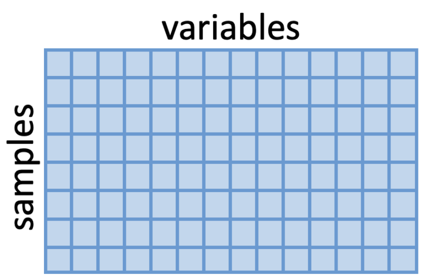

FIGURE 1. A data matrix, samples as rows, variables as columns

Data analysis seeks to uncover insights from such a matrix. What are the principles and processes behind the matrix entries? Finding building blocks for the data can be achieved by fitting a model to the data. The rows of the matrix are (the transposes of) samples of a vector of observed variables $\mathbf{x} = (x_1, \ldots, x_p)^\top$. Writing observed variables as a linear combination of parts means imposing the model $\mathbf{x} = A\mathbf{z}$. At the level of samples, this says $\mathbf{x}^{(i)} = A\mathbf{z}^{(i)}$. Collecting together across all samples, this yields the matrix equation

$$X = ZA^\top,$$

where $X \in \mathbb{R}^{n \times p}$ is the data matrix and $Z \in \mathbb{R}^{n \times q}$ is the scores matrix. The matrix $A \in \mathbb{R}^{p \times q}$ is called the mixing matrix: it encodes how the latent variables combine together to form the observed variables. The above matrix product can equivalently be written as a sum

$$X = \sum_{k=1}^{q} \mathbf{z}_k \mathbf{a}_k^\top,$$

where $\mathbf{z}_k$ and $\mathbf{a}_k$ are the kth columns of Z and A, respectively.

The matrix product is rank q: the p observed variables have been expressed via q factors. By factor, here, we mean a process that describes the data. We don't get to see the latent factors. Instead, we extract (in the rows of Z) how each factor is expressed in a sample and

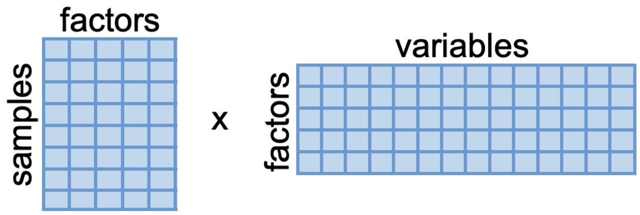

FIGURE 2. A low-rank decomposition $X = ZA^\top$ of a data matrix writes it in terms of factors: the columns of Z record how much the factor is expressed in each sample, while the rows of $A^\top$ record how much each observed variable participates in it. Here there are $n = 8$ samples, $p = 14$ variables, and $q = 5$.

(in the columns of A) how each variable participates in the factor. One thinks of this as *being* the factor: you are where you show up, or you are what you consist of. The collection of variables or samples that participate in a factor, e.g. those whose coefficient is above some threshold, is referred to as a module (not in the algebraic sense of module).

For example (in the table above): gene expression patterns can be encoded via gene modules, brain data via neural sources, images via features, and texts via their concepts and topics.

3.2. Principal component analysis (PCA). PCA is the first example of a model $\mathbf{x} = A\mathbf{z}$. The mixing map A is assumed to be orthogonal: the transformation from $\mathbf{z}$ to $\mathbf{x}$ is an embedding that preserves distances. The latent variables $\mathbf{z} = (z_1, \ldots, z_q)^\top$ are assumed to be uncorrelated. This means there are no linear relationships between the z_i: they are (linearly) separate signals in the data.

DEFINITION 1 (Covariance). *Two variables z_1 and z_2 have* covariance

$$\mathrm{Cov}(z_1, z_2) := \mathbb{E}[z_1 z_2] - \mathbb{E}[z_1]\mathbb{E}[z_2].$$

A collection of variables $\mathbf{z} = (z_1, \ldots, z_q)$ has covariance matrix

$$\Sigma_{\mathbf{z}} := \mathbb{E}[\mathbf{z} \otimes \mathbf{z}] - \mathbb{E}[\mathbf{z}] \otimes \mathbb{E}[\mathbf{z}].$$

This is the $q \times q$ PSD matrix with (i, j) entry $(\mathbb{E}[\mathbf{z} \otimes \mathbf{z}] - \mathbb{E}[\mathbf{z}] \otimes \mathbb{E}[\mathbf{z}])_{ij} = \mathrm{Cov}(z_i, z_j)$. The (i, i) entry is the variance of z_i.

DEFINITION 2 (Uncorrelated). *Two variables z_1 and z_2 are uncorrelated if $\mathrm{Cov}(z_1, z_2) = 0$. A collection of variables $\mathbf{z}$ are mutually uncorrelated if their covariance matrix $\Sigma_{\mathbf{z}}$ is diagonal.*

The relationship $\mathbf{x} = V\mathbf{z}$ implies the following relation between covariance matrices:

$$\Sigma_{\mathbf{x}} = V \Sigma_{\mathbf{z}} V^\top,$$

where the matrices $\Sigma_{\mathbf{z}}$ and $\Sigma_{\mathbf{z}}$ are the covariance matrices of $\mathbf{x}$ and $\mathbf{z}$, respectively. The covariance $\Sigma_{\mathbf{z}}$ is diagonal, since the entries of $\mathbf{z}$ are uncorrelated. This factorization write a real symmetric matrix as the product of orthogonal times diagonal times orthogonal transpose: it is the eigendecomposition. Hence the factors V and $\Sigma_{\mathbf{z}}$ exist and are generically unique, by the spectral theorem.

The matrix $\Sigma_{\mathbf{x}}$ is the population covariance of the observed variables. In practice, we do not have access to the population covariance and instead use the sample covariance matrix:

DEFINITION 3 (Sample covariance). *Given data* $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)} \in \mathbb{R}^p$ *with sample mean* $\bar{\mathbf{x}} = \frac{1}{n}\sum_i \mathbf{x}^{(i)}$, *the* sample covariance matrix *is*

$$S = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}^{(i)} - \bar{\mathbf{x}})(\mathbf{x}^{(i)} - \bar{\mathbf{x}})^\top.$$

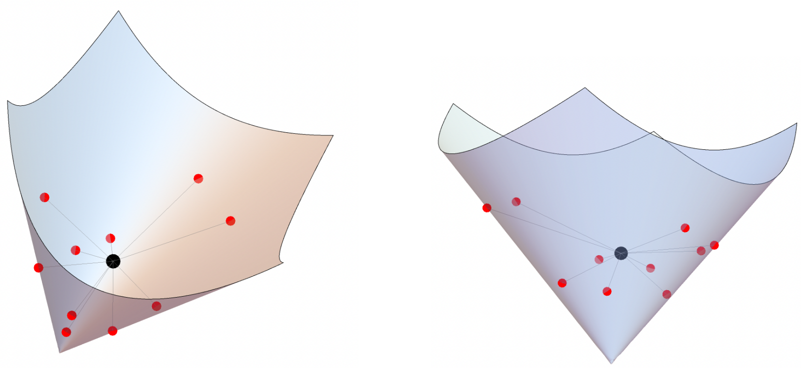

FIGURE 3. Covariance matrices are positive semi-definite (PSD). Here is a picture of the 2×2 PSD cone (two views). Its boundary consists of rank one PSD matrices; i.e., $\{\mathbf{v}\mathbf{v}^\top : \mathbf{v} \in \mathbb{R}^2\}$. Here 10 points (say $\mathbf{v}\mathbf{v}^\mathsf{T}$ for $\mathbf{v} \in \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\}$) on the boundary are shown in red. Their average, after mean-centering, is the sample covariance matrix of the data $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}$. It is shown here in black.

What does PCA do?

- It truncates the eigendecomposition of the sample covariance matrix to some rank r: it keeps the top r eigenvalues (and their corresponding eigenvectors) and sets the rest to 0.
- Equivalently, it finds the best rank r approximation of the sample covariance matrix, where this equivalence holds by the Eckart-Young theorem.
- Equivalently, it finds the singular value decomposition of the mean-centered data matrix X. That is, it finds orthogonal matrices U and V and a diagonal matrix D such that $X = UDV^\mathsf{T}$. The eigendecomposition and SVD formulations are equivalent, because
$$\frac{1}{n}X^\mathsf{T}X = \frac{1}{n}(UDV^\mathsf{T})^\mathsf{T}(UDV^\mathsf{T}) = VD^2V^\mathsf{T}.$$

The eigenvector corresponding to the ith eigenvalue is the ith principal component (PC). This is well-defined if the top r eigenvalues of the sample covariance matrix are distinct. Having computed the PCs, PCA then projects the data points $\mathbf{x}^{(i)} \in \mathbb{R}^p$ to the r-dimensional space spanned by the r principal components.

REMARK 4 (Taking the transpose). *From a linear algebra point of view, the SVD of a matrix* $X = UDV^\top$ *yields the SVD of its transpose* $X^\top = VDU^\top$. *From a data analysis point of view, the columns of* U *and* V *are thought of differently: one records the extent to which each factor appears in each sample, while the other records how each variable participates in the factor. When plotting data in their new space, one matrix gives the coordinates of the samples in the new space, while the other gives the axes in the space.*

3.3. Tensor decomposition. Above, our data consisted of points $\mathbf{x}^{(i)}$ in $\mathbb{R}^p$. Many datasets, though, carry multi-indexed structure, either on their space of variables, space of samples, or both. Examples include: fluorescence spectroscopy (samples $\times$ emission wavelengths $\times$ excitation wavelengths), color images (height $\times$ width $\times$ RGB channel), or neural recordings (neuron $\times$ experiment $\times$ time). See [8] for these and other examples. As with a data matrix, one can decompose a data tensor into rank one pieces. Each summand is a factor. A factor 'is' its expression in each sample, the amount each variable participates in it, and its presence across each context.

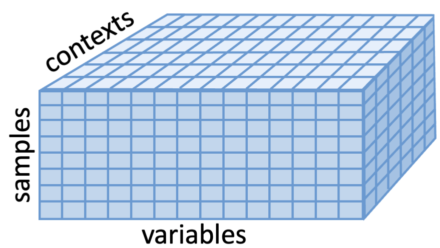

FIGURE 4. A data tensor, samples as rows, variables as columns, contexts as tubes.

Theoretical and practical questions remain unanswered for optimal extraction of interpretable factors from tensors of data (i.e. for low rank approximation of tensors). For example: how should decompositions of different ranks be compared or combined? For PCA/SVD, the solution as truncation of the eigendecomposition to the top r eigenvalues mean the optimal solutions have a natural nested structure. This doesn't happen in general for tensors.

3.4. Independent component analysis (ICA). Independent Component Analysis (ICA) is a related model to PCA; it is also of the form $\mathbf{x} = A\mathbf{z}$, where the variables $\mathbf{x}$ are oberved, the variables $\mathbf{z}$ are latent, and the mixing matrix A relates them. ICA makes two changes compared to PCA: it strengthens the uncorrelated assumption on $\mathbf{z}$ to independence, and it removes the orthogonality restriction on A.

DEFINITION 5. *Variables $z_1, \ldots, z_q$ are* independent *if the joint probability density of $\mathbf{z} = (z_1, \ldots, z_q)^\top$ factors as $f_{\mathbf{z}}(t_1, \ldots, t_q) = f_{z_1}(t_1) \cdots f_{z_q}(t_q)$.*

Independence implies uncorrelated, but it is stronger: two variables being uncorrelated ruled out *linear* dependences between them; two variables being independent rules out *any* dependence between them: conditioning on the value of one variable doesn't change the density of the other.

The classical motivating example for ICA is the *cocktail party problem*: p microphones each record a mixture $x_i = \sum_j A_{ij} z_j$ of independent sound sources z_j; the goal is to recover A and the sources from the microphone recordings. This is an example of a blind source separation problem.

One can recover the parameters in an ICA model via tensor decomposition of cumulant tensors [6], as we now describe.

DEFINITION 6 (Moments). *The d-th* moment tensor *of $\mathbf{z}$ is $\mathcal{M}_d(\mathbf{z}) := \mathbb{E}[\mathbf{z}^{\otimes d}] \in S^d(\mathbb{R}^q)$ with entries $\mathcal{M}_d(\mathbf{z})_{i_1 \ldots i_d} = \mathbb{E}[z_{i_1} \cdots z_{i_d}]$.*

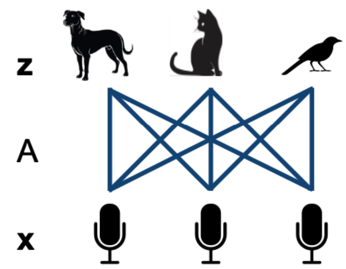

FIGURE 5. ICA for the cocktail party problem. Sources $\mathbf{z}$ are mixed together via mixing matrix A to produce observed variables $\mathbf{x}$.

The first moment is the mean: $\mathcal{M}_1(\mathbf{z}) = \mu$; the second moment is the matrix $\mathcal{M}_2(\mathbf{z}) = \mathbb{E}[\mathbf{z} \otimes \mathbf{z}]$, which relates to the covariance by $\Sigma_{\mathbf{z}} = \mathcal{M}_2(\mathbf{z}) - \mu \otimes \mu$, as we saw above.

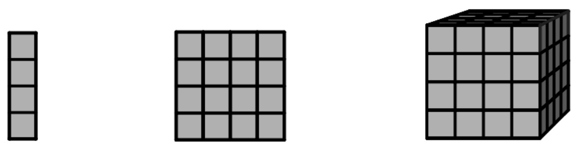

FIGURE 6. The first moment is a vector, the second is a matrix, the third is a third order tensor.

DEFINITION 7 (Moments and cumulants). *For* $\mathbf{t} = (t_1, \ldots, t_q)^\mathsf{T}$*, the* moment generating function *(mgf) of* $\mathbf{z}$ *is* $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp\langle \mathbf{t}, \mathbf{z}\rangle]$*, and the* cumulant generating function *(cgf) is* $K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t})$*. We have*

$$M_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 0} \frac{1}{d!} \big\langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z}) \big\rangle.$$

*and define the d-*th cumulant tensor $\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$ *as the analogous coefficient of* $K_{\mathbf{z}}$*:*

$$K_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 1} \frac{1}{d!} \big\langle \mathbf{t}^{\otimes d}, \mathcal{K}_d(\mathbf{z}) \big\rangle.$$

PROPOSITION 8 (The cumulant tensors of independent variables are diagonal). *If* $z_1, \ldots, z_q$ *are independent, then* $\mathcal{K}_d(\mathbf{z})$ *is diagonal for every* d*.*

PROOF. Independence gives $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}\big[\exp(t_1 z_1) \cdots \exp(t_q z_q)\big] = \prod_{i=1}^q \mathbb{E}[\exp(t_i z_i)]$, so

$$K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t}) = \sum_{i=1}^{q} \log \mathbb{E}[\exp(t_i z_i)],$$

which is a sum of univariate functions. Hence the degree-d coefficient tensor $\mathcal{K}_d(\mathbf{z})$ has entries supported only on indices $(i, i, \ldots, i)$. $\square$

DEFINITION 9 (The action of A on a symmetric tensor). *For* $A \in \mathbb{R}^{p \times q}$ *and* $S \in S^d(\mathbb{R}^q)$*, define* $A \bullet S \in S^d(\mathbb{R}^p)$ *by*

$$(A \bullet S)_{i_1 \ldots i_d} = \sum_{\alpha_1, \ldots, \alpha_d = 1}^{q} S_{\alpha_1 \ldots \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d}.$$

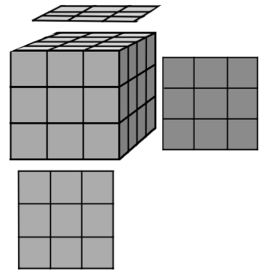

FIGURE 7. The action $A \bullet S$ multiplies a tensor by a matrix along each index.

For $d = 2$, $(A \bullet S)_{ij} = \sum_{\alpha\beta} S_{\alpha\beta} A_{i\alpha} A_{j\beta} = (ASA^\mathsf{T})_{ij}$: this is matrix congruence.

PROPOSITION 10 (Multilinearity). *Let* $\mathbf{x} = A\mathbf{z}$ *for* $A \in \mathbb{R}^{p\times q}$. *Then* $\mathcal{M}_d(\mathbf{x}) = A \bullet \mathcal{M}_d(\mathbf{z})$ *and* $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z})$.

PROOF.

$$\mathcal{M}_d(\mathbf{x})_{i_1\dots i_d} = \mathbb{E}[x_{i_1} \cdots x_{i_d}] = \mathbb{E}\Big[\prod_{k=1}^{d} \sum_{\alpha_k} A_{i_k\alpha_k} z_{\alpha_k}\Big] = \sum_{\alpha_1,\dots,\alpha_d} A_{i_1\alpha_1} \cdots A_{i_d\alpha_d}\, \mathbb{E}[z_{\alpha_1} \cdots z_{\alpha_d}],$$

using linearity of expectation. The last factor is $\mathcal{M}_d(\mathbf{z})_{\alpha_1\dots\alpha_d}$, giving $(A \bullet \mathcal{M}_d(\mathbf{z}))_{i_1\dots i_d}$. The same argument applies to the coefficient tensors from the cumulant generating function. $\square$

PROPOSITION 11 (ICA is symmetric tensor decomposition). *Fix* $d \geq 3$. *Assume* $\mathbf{x} = A\mathbf{z}$ *where the variables* $\mathbf{z}$ *are mutually independent. Writing* $\mathbf{a}_i$ *for the ith column of* A *and* $\lambda_i := \mathcal{K}_d(\mathbf{z})_{i,i,\dots,i}$ *(the dth cumulant of the ith source), then we have*

$$\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^{p} \lambda_i\, \mathbf{a}_i^{\otimes d}.$$

PROOF. The cumulant tensors of mutually independent variables are diagonal. By the cumulant analog of multilinearity of moments (Proposition 10), $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_i \lambda_i\, (A \bullet \mathbf{e}_i^{\otimes d})$. We have $(A \bullet \mathbf{e}_i^{\otimes d})_{j_1\dots j_d} = A_{j_1 i} \cdots A_{j_d i} = (\mathbf{a}_i^{\otimes d})_{j_1\dots j_d}$, so $A \bullet \mathbf{e}_i^{\otimes d} = \mathbf{a}_i^{\otimes d}$. $\square$

Methods to analyze data give factorizations. In this section, we saw two examples: (i) PCA and the eigendecomposition, and (ii) ICA and symmetric tensor decomposition. A third classical example, omitted from these notes, is the connection between LDL decompositions (or the Cholesky or QR decompositions) and linear structural equation model.

4. The hiking trail

We call the previous section the paved trail because it is a canonical and well-established connection between data analysis and (multi-)linear algebra. This section turns to other methods and their factorizations. These are a hiking trail rather than a paved trail: the route exists, but it is less well-traveled. In the A4 lectures, we'll talk about how one can use these factorizations to tackle new data analysis questions.

So far we have looked at a single set of observed variables $\mathbf{x} = (x_1, \dots, x_p)$. In many settings, one has a finer structure: a set of variables $\mathbf{x}$ observed under multiple contexts (also called views or environments).

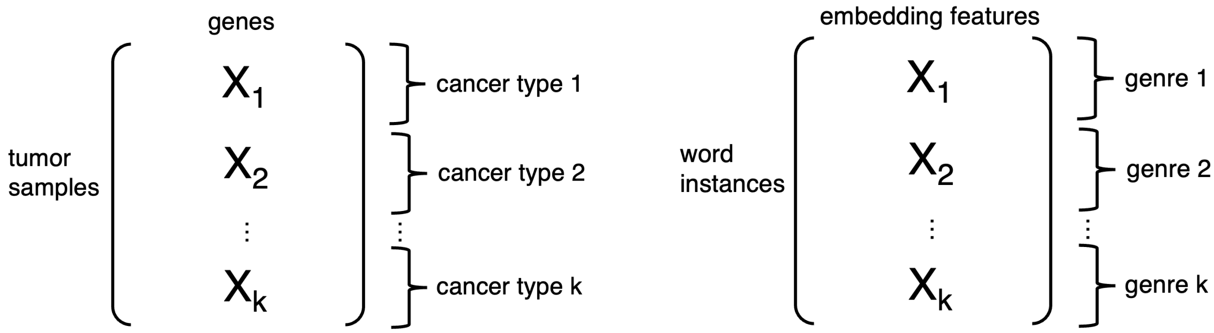

FIGURE 8. Two examples of datasets in which the variables are observed under multiple contexts. The dataset X_i in the ith context is an $n_i \times p$ matrix, where n_i is the number of samples in context i. The total number of samples is $n_1 + \cdots + n_k$. The same p variables are measured across all k contexts. The left picture shows genes whose expression levels are recorded in tumor samples. Each tumor sample has a cancer type. The right shows features of different words, obtained by embedding via an LLM. Each word instance appears in a text of some genre.

Sticking to the paved trail, two options present themselves: (i) analyze the data in each context separately, and combine the insights at the end, (ii) combine all the data together, ignoring the contextual information. A third option, which we pursue here, attempts the best of both: make shared and individual insights from the datasets.

The idea of principal components that are shared across multiple datasets originates in work of Flury in the 1980s [7]. The *hypothesis of common principal components* says that the covariance matrices from the different contexts share the same principal components but their weights vary (they have the same eigenvectors but different eigenvalues). Letting Σ_i be the covariance matrix of the ith dataset, the model says that

$$\Sigma_i = V D_i V^{\mathsf{T}}, \tag{1}$$

where $V \in \mathbb{R}^{p\times p}$ is the matrix of eigenvectors and $D_i \in \mathbb{R}^{p\times p}$ the corresponding eigenvalues in context j. In practice, as in usual PCA, the eigendecompositions are truncated to the top r, to give $V \in \mathbb{R}^{p\times r}$ and $D_i \in \mathbb{R}^{r\times r}$.

The study of multiple datasets replaces diagonalization (of a single covariance matrix, data tensor, or cumulant tensor) with simultaneous diagonalization (of a collection of matrices or tensors). The link between simultaneous diagonalization and tensor decomposition is well known [4]. Stacking k covariance matrices, each on the same set of p variables, builds a $p \times p \times k$ tensor T with $T_{\alpha\beta i} = (\Sigma_i)_{\alpha\beta}$; see below. Decomposition of this tensor is a simultaneous diagonalization problem.

PROPOSITION 12. *The partially symmetric tensor decomposition*

$$T = \sum_{j=1}^{r} \mathbf{v}_j^{\otimes 2} \otimes \mathbf{d}_j, \tag{2}$$

is equivalent to the coupled matrix decompositions in (1)*, where* $\mathbf{v}_j \in \mathbb{R}^p$ *are the columns of* V *and where* $\mathbf{d}_j = ((D_1)_{jj}, \ldots, (D_k)_{jj})) \in \mathbb{R}^k$.

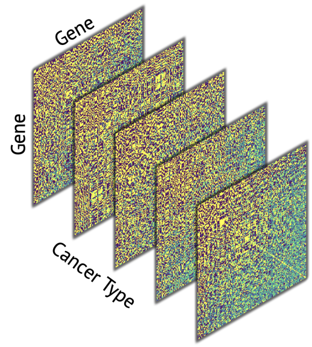

FIGURE 9. A stack of $p \times p$ covariance matrices, one for each of k contexts, forms a $p \times p \times k$ tensor. In this picture, the co-variation between $p = 500$ genes are studied across and $k = 5$ cancer types. This is the stack of covariance matrices resulting from the data on the left of Figure 8.

PROOF. The decomposition in (2) says $T_{\alpha\beta i} = \sum_{j=1}^{r} (\mathbf{v}_j)_\alpha (\mathbf{v}_j)_\beta (\mathbf{d}_j)_i = \sum_{j=1}^{r} V_{j\alpha} V_{j\beta} (D_i)_{jj} = (V D_i V^\mathsf{T})_{\alpha,\beta}$, so $T_{:,:,i} = V D_i V^\mathsf{T}$. □

For a small enough rank r, this decomposition is identifiable: the factors $\mathbf{v}_i$ and $\mathbf{d}_i$ are unique up to scale and permutation, see e.g. [5]. Orthogonality of the $\mathbf{v}_i$ is not required for identifiability.

Partially symmetric tensor decompositions are well studied from an algebraic perspective, especially in the case of complex-valued vectors [2,9]. It is a topic of ongoing work to use this algebraic insight for practical, robust algorithms to compute them. Our recent work on multi-context PCA (MCPCA) [12] is an algorithm for partially symmetric tensor decomposition, to find axes of variation in data and the contexts in which they appear. An added feature of the decomposition above is its partial non-negativity: the entries of the vectors $\mathbf{d}_j$ are interpreted as the extent to which direction $\mathbf{v}_i$ participates each context. The weights of these contributions should be non-negative, by analogy to the eigenvalues of the covariance matrix. To achieve this, the algorithm for MCPCA combines tensor decomposition with non-negative least squares.

Above we have considered the case of a general number of contexts. The case of two contexts ($k = 2$) is of particular interest. The two datasets could represent, for example, a patient group for a disease and a group of healthy controls. The goal is to identify structure present in a foreground dataset after removing the effect of the background. Several methods have been devised for this purpose. The joint diagonalization of a pair of matrices is given by the generalized SVD [3, 11]. Other methods include contrastive PCA [1] and contrastive ICA [13].

REMARK 13 (Taking the transpose, part 2). *Remark 4 explored the data analysis meaning behind taking the transpose the SVD of a data matrix. In a similar vein, we can imagine taking the transpose of the block matrices in Figure 8. Then, instead of a set of variables measured across different contexts, we have a set of samples measured in multiple data modalities.*

5. Going off trail

Problems across data science can be viewed as matrix and tensor factorization problems. These aren't necessarily factorizations that have received an algebraic treatment before. The previous section gave one example: a decomposition that is both partially symmetric and partially non-negative.

There are many unexplored directions. Here are a few that branch off our hiking trail; i.e., that concern extensions of MCPCA. Above, the contexts were indexed as discrete categories $\{1, \ldots, k\}$, but in practice the contexts may be ordered, continuous, hierarchical, unknown, overlapping, nested, etc. Which new tensor decompositions will we uncover in these settings? What will their decompositions, identifiability, and optimization landscapes look like?

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